Ajay Arvind Chodankar

Data Scientist | AI Engineer | Software Engineering | Cloud & Automation
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CAREER SUMMARY

Software engineer focused on applied ML and platform engineering. I build LLM-powered services end to end: data ingestion, retrieval (RAG), evaluation, and deployment (Docker, CI/CD, Kubernetes). Recent work improved workflow accuracy to ~90% and reduced deployment time by up to 90% through automation.
Languages – English (C1), German (B1), and Hindi (Native)

EDUCATION

MASTER'S DEGREE | Data Science | **Technische Universität Braunschweig**, DE **2022 – 2025**
Grade – 1.9
BACHELOR'S DEGREE | Computer Engineering | **University of Mumbai**, IN **2018 – 2022**
Grade – 8.75/10 CGPA (1.6 German Equivalent)

CORE AREAS

Generative AI (LLMs, RAG) | NLP | Data Engineering | Cloud/DevOps Automation | Microservices | Computer Vision | Simulation & Autonomous Driving | AEC/BIM Analytics

TECHNICAL SKILLS

GenAI / NLP – LLMs | Prompt engineering | RAG | LangChain | Evaluation/QA | Multi-agent orchestration (CrewAI)
Programming / Data – Python | C# | Rust | SQL | FastAPI | ETL pipelines | Data analysis | Statistics
Cloud / DevOps / Tools – Docker | Kubernetes | GitLab CI/CD | Ansible | AWS | Azure | GCP | Grafana | Tableau | CARLA

PROFESSIONAL EXPERIENCE

RESEARCH ASSISTANT - DevOps & System Administrator **Apr 2024 – Present**
TU Braunschweig | Braunschweig, DE
Manage lab and project infrastructure end to end, from provisioning and build standardization to automated deployments and day-to-day Linux/Windows reliability.

- **Cut infrastructure deployment time by ~70%** by standardizing provisioning with Ansible (IaC) across Ubuntu, Rocky Linux, and Windows; reduced manual setup effort by ~80%.
- **Automated recurring operations** (DB updates, internal site refreshes, housekeeping) with Python scripts, reducing manual intervention and errors.
- **Hardened delivery workflows** with GitLab CI/CD and Docker, adding automated tests and repeatable builds for safer, more frequent releases.

RESEARCH ASSISTANT - AI/ML ENGINEER **Jun 2024 – Feb 2026**
TANGILITY GmbH / TU Braunschweig | Braunschweig, DE
Build and ship ML features for AEC/VR product. Deliver retrieval-backed assistants, geometry/quality checks, and containerized APIs that integrate with existing CAD/Unity workflows.

- **Improved LLM response quality and consistency** by integrating Azure AI Foundry APIs, iterating prompts with test sets, and adding lightweight evaluation/QA checks.
- **Built a CAD-to-Unity VR automation pipeline** (Python, FastAPI, Docker) that runs vision-based object recognition and rule-based mapping to assign real-world properties and auto-attach Unity scripts, producing interactive scenes instead of static imports.
- **Detected early-stage design conflicts** by analyzing collision/geometry data and generating reports that surfaced spatial issues earlier in the project lifecycle.

GRADUATE APPRENTICE - DATA ANALYST **Jun 2022 – Aug 2022**
LTI - Larsen & Toubro Infotech | Mumbai, IN
Hands-on project work in data engineering and cloud. Built batch/stream pipelines and dashboards, and validated results with basic ML and statistical checks.

- **Developed automated ETL pipelines** using PySpark, Kafka, and Hadoop to simulate high-volume data streaming and real-time processing.
- **Executed cloud data engineering workflows** on GCP and AWS, managing the end-to-end ingestion and cleaning of large-scale structured and unstructured datasets.
- **Built interactive Tableau dashboards** and applied ML algorithms to validate statistical models, transforming raw lab data into actionable business insights.

PROJECTS

Cura – Learn to Care

AI-assisted learning platform for nursing education—organizes course material and lets students study with grounded Q&A and practice tools.

- **Built the core web platform** (Flask, FastAPI, SQLAlchemy) with onboarding, course management, quizzes, and teacher uploads.
- **Implemented RAG-based study features** using Qdrant and document ingestion so students could ask questions and generate quizzes/flashcards from uploaded PDFs.
- **Added voice + avatar interaction** with speech-to-text/text-to-speech for more accessible practice and guided explanations.

GitHub: TheCez/cura_ai-learning-support

Autonomous Valet Parking Path Planning and Conflict Resolution System (Master Thesis)

Built an autonomous valet-parking simulation and planning stack with collision-aware routing and a clean simulator-to-planner interface.

- **Connected CARLA to CommonRoad** to convert live simulation scenes into structured planning problems with routes and parking goals.
- **Built real-time state sharing and conflict detection** (occupancy grid, multi-vehicle sync) using ZeroMQ to flag potential path collisions and coordinate maneuvers.
- **Implemented Hybrid A* planning and vehicle control** (PID), plus logging and dynamic replanning; explored ML ideas for waypoint prediction and conflict resolution.

GitHub: TheCez/commonroad_path_planner

CERTIFICATIONS

- **Generative AI Engineering with LLMs Specialization -** IBM · Apr 2026
- **LBJ Big Data Bootcamp -** LTI · Jul 2022 · Hadoop, Spark, PySpark, Kafka
- **Microsoft Azure Fundamentals (AZ-900) -** Microsoft · Mar 2022

ACHIEVEMENTS

- **Solves a BIG AEC Problem -** AEC Hackathon Munich 2024
- **Best LLM Idea -** Healthcare Hackathon 2024 held in Berlin (UKSH)
- **Winner Rank 1 -** HealthHack 2024 held in Braunschweig

PUBLICATIONS

- V. Fegade, A. Chodankar, A. Bhingle and S. Mhatre, "Residential Security System based on Facial Recognition," 2022 6th International Conference on Trends in Electronics and Informatics (ICOEI), Tirunelveli, India, 2022, pp. 01-09, doi: 10.1109/ICOEI53556.2022.9776940. keywords: {Performance evaluation;Privacy;Costs;Face recognition;Heuristic algorithms;Lighting;Real-time systems;face recognition;person identification;convolutional neural networks;machine learning;deep learning;biometrics;SSD;BlazeFace;MobileFaceNets;Euclidean Distance algorithm},

LANGUAGES

- English (C1)
- German (B1)
- Hindi (Native)

Braunschweig, 22 April 2026
City, date

Ajay Arvind Chodankar